



Customer Shopping Behavior Analysis

Uncovering insights from 3,900 purchases to guide strategic business decisions

DATASET OVERVIEW

The Foundation

3,900

Total Purchases

Transactions analyzed
across all categories

18

Data Points

Features tracked per
customer

50

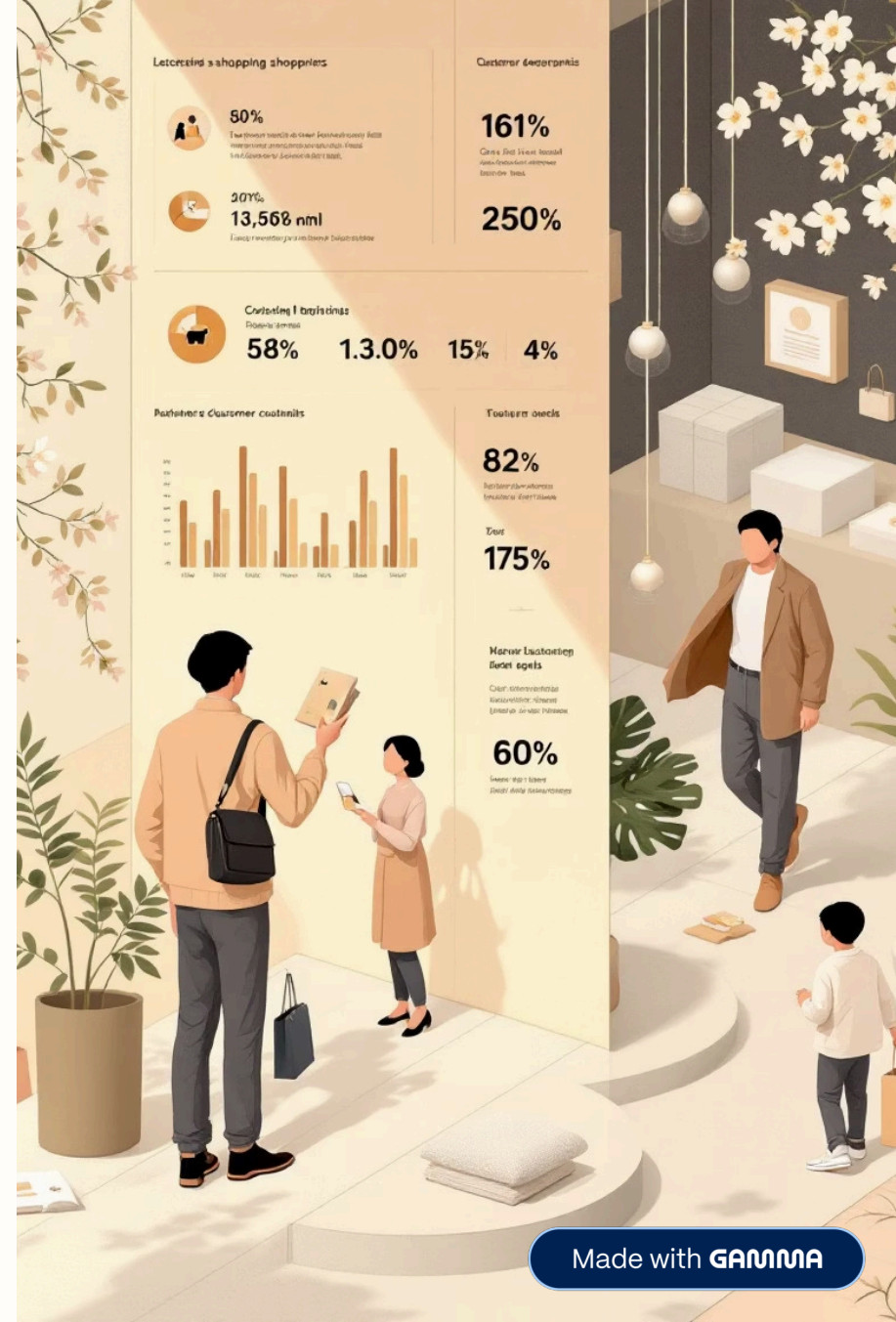
Locations

Geographic coverage

4

Categories

Product segments



Data Preparation & Cleaning

01

Data Loading

Imported dataset using pandas

02

Exploration

Structure check and summary statistics

03

Missing Data

Imputed 37 missing review ratings by category median

04

Feature Engineering

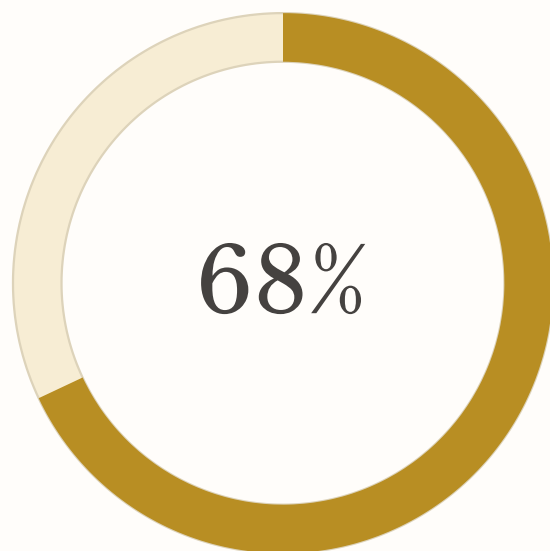
Created age groups and purchase frequency metrics

05

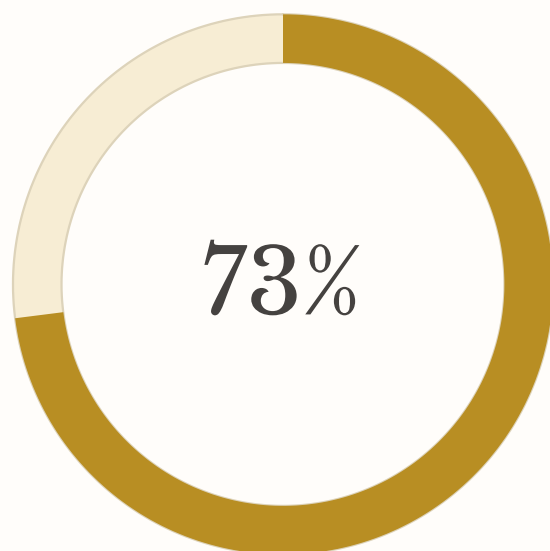
Database Integration

Loaded cleaned data into PostgreSQL

Customer Profile Snapshot



Male Customers



Non-Subscribers

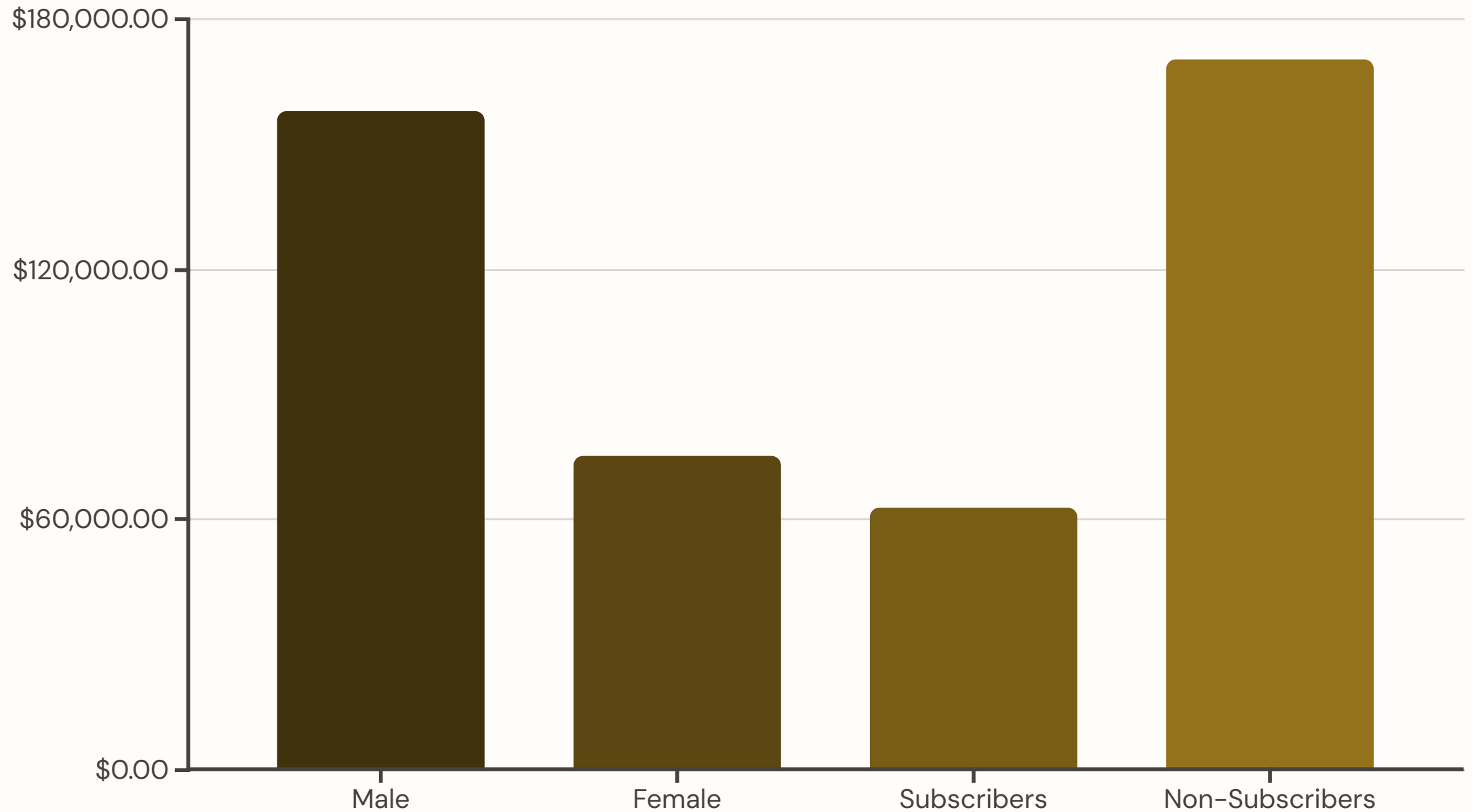
Demographics

- Average age: 44 years
- Age range: 18–70 years
- Most common size: Medium

Purchase Behavior

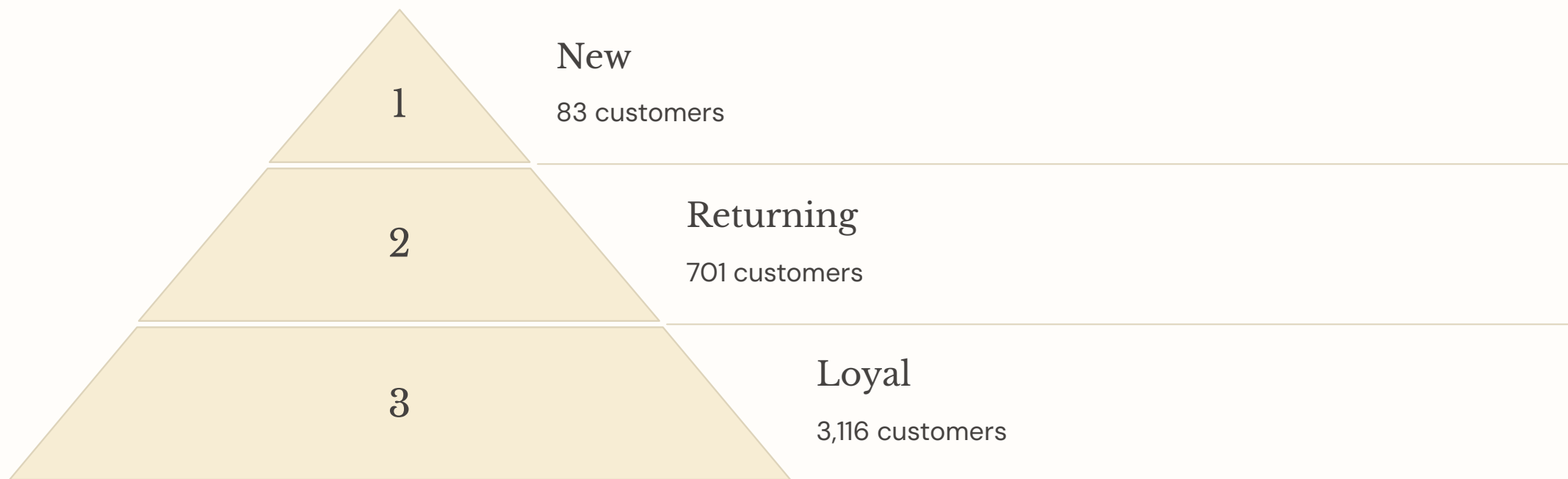
- Average spend: \$59.76
- Average rating: 3.75/5

Revenue Analysis



Key Finding: Male customers generate 2x more revenue than female customers, while non-subscribers contribute significantly more total revenue despite lower average spend.

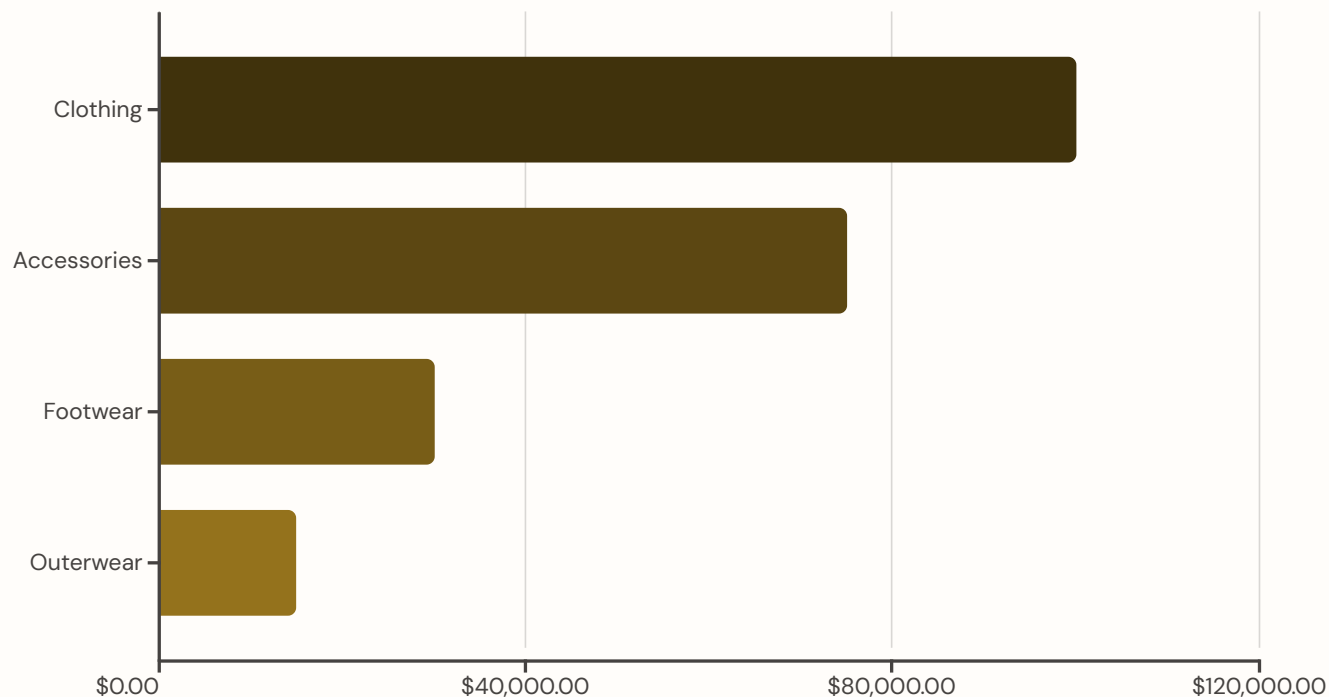
Customer Segmentation



80% of customers are loyal based on purchase history, presenting strong retention opportunities

Top Products & Categories

Revenue by Category



Top Rated Items

1. Gloves (3.86)
2. Sandals (3.84)
3. Boots (3.82)
4. Hat (3.80)
5. Skirt (3.78)

📌 Clothing dominates with 45% of total revenue



Discount & Shipping Insights

High-Value Discount Users

839 customers used discounts but spent above average

Discount-Dependent Products

Hat (50%), Sneakers (49.7%), Coat (49.1%)

Shipping Preference

Express shipping: \$60.48 avg vs Standard: \$58.46 avg



 POWER BI

Interactive Dashboard



Customer Metrics

3.9K customers, \$59.76 avg purchase,
3.75 rating



Dynamic Filtering

By subscription, gender, category,
shipping type



Visual Analytics

Revenue trends, age groups, category
performance

Strategic Recommendations



Boost Subscriptions

Promote exclusive benefits to convert 73% non-subscribers



Loyalty Programs

Reward repeat buyers to strengthen 80% loyal base



Optimize Discounts

Balance sales boosts with margin control



Targeted Marketing

Focus on high-revenue segments and express shipping users



Product Positioning

Highlight top-rated items in campaigns